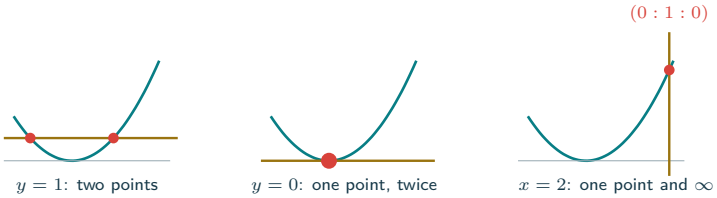
PROJECTIVE GEOMETRY LADDER 4 OF 9

Bézout's theorem and multiplicity

With infinity added and every root available, curves of degrees m and n meet in exactly mn points.

1 / THE RULE AND THE PICTURE

Two plane curves of degrees m and n with no common component meet in exactly mn points



$y = 1$: two points $\quad$ $y = 0$: one point, twice $\quad$ $x = 2$: one point and ∞

The count is made in the projective plane over an algebraically closed field, with each point counted by its multiplicity:

$$\#(C \cap D) = m\,n$$

A line meets the parabola twice in every picture, though the second point may be double, at infinity, or complex.

2 / EXAMPLES WITH THE COUNT

Every pair of curves reaches its full count

curves (degrees)	intersection points	count
$y = 1$ and $y = x^2$ $(1 \cdot 2)$	$(1, 1)$ and $(-1, 1)$	2
$y = 0$ and $y = x^2$	$(0, 0)$, a double root	2
$x = 2$ and $y = x^2$	$(2, 4)$ and $(0 : 1 : 0)$	$1 + 1$
$y = -1$ and $y = x^2$	$(i, -1)$ and $(-i, -1)$	2, complex
$x^2 + y^2 = 1$ and $y = x^2$ $(2 \cdot 2)$	$y = \frac{-1 \pm \sqrt{5}}{2}$, so $x = \pm 0.786$ and $x = \pm 1.272\,i$	$2 + 2$
$x^2 + y^2 = 1$ and $x^2 + y^2 = 4$	none, but $(1 : \pm i : 0)$ each twice	$2 + 2$
$y = 0$ and $y = x^3$ $(1 \cdot 3)$	$(0, 0)$, a triple root	3
$x + y = 1$ and $x + y = 3$	$(-1 : 1 : 0)$	1

3 / FOUR THINGS THE THEOREM NEEDS

Remove any one and the count fails

Without the line at infinity the lines $x + y = 1$ and $x + y = 3$ have no common point, and the line $x = 2$ meets the parabola only once.
Without an algebraically closed field the line $y = -1$ misses the parabola over the reals, since $x^2 = -1$ has no real root.
Without multiplicity the tangent $y = 0$ meets the parabola once and not twice, and without the condition on components two copies of the same line meet in infinitely many points.

4 / HOW MULTIPLICITY IS COUNTED

The number of times a root repeats

Substitute one curve into the other and count repeated roots: putting $y = 0$ into $y = x^3$ gives $x^3 = 0$, a triple root, so the line and the cubic meet at the origin with multiplicity 3, which is the full count $1 \cdot 3$ for the cubic's flex.
For two circles with equal centres the homogeneous equations $X^2 + Y^2 = Z^2$ and $X^2 + Y^2 = 4Z^2$ subtract to $3Z^2 = 0$, so $Z = 0$ twice, and each of the two circular points $(1 : \pm i : 0)$ carries multiplicity 2, so the circles are tangent to each other at these two points.

5 / CONSEQUENCES

Counting lets us determine curves by points

A conic has six coefficients up to a common factor, so five points in general position fix a conic, and two different conics share at most 4 points.
A line meets a cubic in 3 points, so a line through two rational points of a cubic meets it in a third point that is again rational, which the next sheets use.

6 / TRY THESE

Count the intersection

(a) How many points do two conics share? Four, counting complex points, points at infinity and multiplicity.
(b) A line and the hyperbola $xy = 1$: $y = 1$ gives $x = 1$ and the point $(1 : 0 : 0)$ at infinity, so $1 + 1 = 2$.
(c) Two parabolas $y = x^2$ and $y = 2x^2$ meet at the origin and at infinity: setting $YZ = X^2$ and $YZ = 2X^2$ gives $X^2 = 0$, so $X = 0$ twice, and $YZ = 0$ gives $(0 : 1 : 0)$ and the origin; each is counted twice for a total of 4.

ASIDE / WHY mn

The product counts the roots of one polynomial of degree mn in a single variable after eliminating the other, which is what the resultant does.

LINKS TO OTHER SHEETS

The previous sheet writes conics and cubics homogeneously; the sheet on chords and tangents uses the count 3 for a line and a cubic.

Convention. m, n are the degrees; multiplicity is the number of repeated roots at a point; the circular points are $(1 : \pm i : 0)$; algebraically closed means every polynomial has a root.